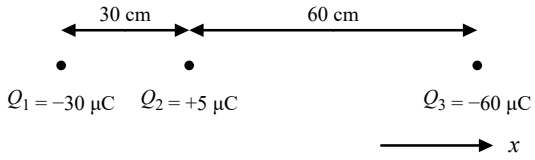
STPM 2013 PHYSICS 960/2

Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}$ C, $N_A = 6.02 \times 10^{23}\ mol^{-1}$, $k = \frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9\ N\ m^2\ C^{-2}$, $\mu_0 = 4\pi \times 10^{-7}\ H\ m^{-1}$. *The Section A answers are worked from the questions.*

Section A [15 marks]

1. Three charged particles which are arranged in a straight line is shown in the diagram below.



What is the electrostatic force acting on Q_3 due to the other two charges?

A 12.5 N in the negative-x direction
B 12.5 N in the positive-x direction
C 27.5 N in the negative-x direction
D 27.5 N in the positive-x direction

Answer: B

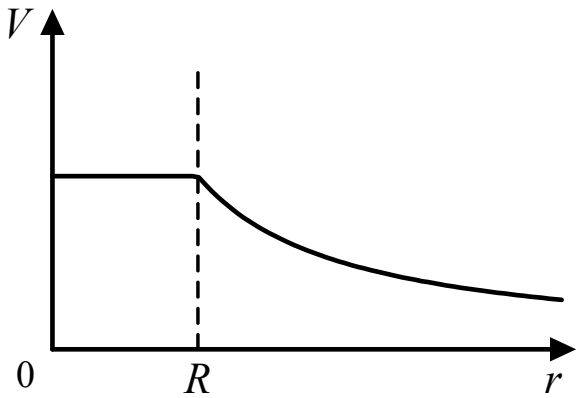
From Q_1 (0.90 m, both negative, repulsion, $+x$): $\frac{(9.0 \times 10^9)(30 \times 10^{-6})(60 \times 10^{-6})}{(0.90)^2} = 20.0$ N

From Q_2 (0.60 m, attraction, $-x$): $\frac{(9.0 \times 10^9)(5 \times 10^{-6})(60 \times 10^{-6})}{(0.60)^2} = 7.5$ N

Resultant = 20.0 − 7.5 = 12.5 N in the positive-x direction (B).

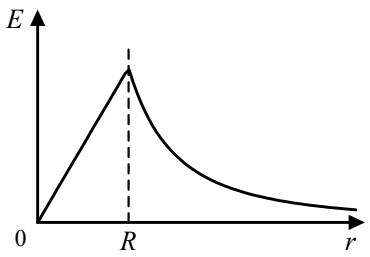
Explanation: Like charges repel, unlike charges attract.

2. The variation of the electric potential V with distance r from the centre of a charge conducting sphere of radius R is shown by the graph below.

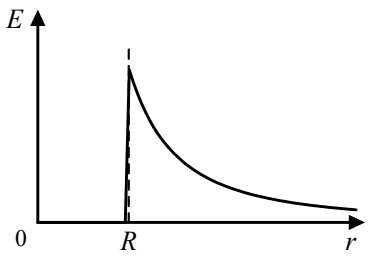


Which graph shows the relationship between the electric field strength E with the distance r?

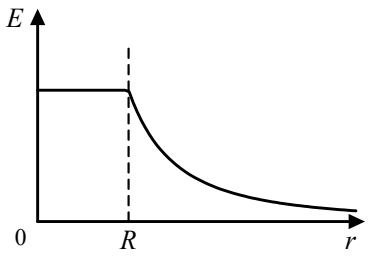
A



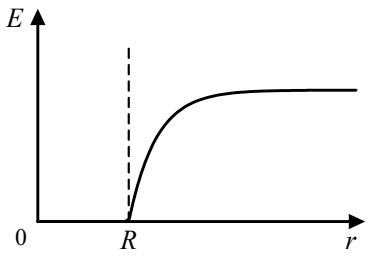
B



C



D



Answer: B

$E = -\frac{dV}{dr}$: V is constant inside, so $E = 0$ for $r < R$; outside E jumps to its maximum at R and falls as $1/r^2$ (B)

Explanation: The field is the negative gradient of the potential.

3. Three capacitors of equal capacitance C can be arranged so as to obtain the minimum capacitance or the maximum capacitance. If the minimum capacitance is P and the maximum capacitance is Q, the ratio of $\frac{P}{Q}$ is

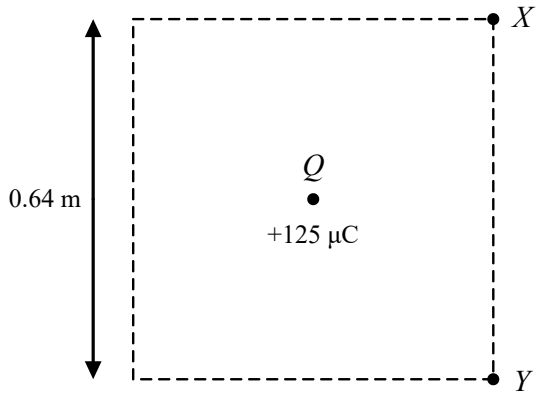
A $\frac{1}{9}$ B 9 C $\frac{1}{C^2}$ D C^2

Answer: A

Minimum (series): $P = \frac{C}{3}$; maximum (parallel): $Q = 3C$; $\frac{P}{Q} = \frac{1}{9}$

Explanation: The ratio is a pure number.

4. A charge Q of +125 μC is fixed at the centre of a square of length 0.64 m is shown in the diagram below.



If a charge P of +7.0 μC is moved from X to Y, what is the work done to the charge Q?

A 0 J B 12.3 J C 17.5 J D 24.6 J

Answer: A

X and Y are corners of the square, the same distance from Q, so they are at the same potential.

$$W = q(V_Y - V_X) = 0 \text{ J}$$

Explanation: The work done between two points depends only on the potential difference.

5. The increase in the electrical resistivity of a metallic conductor when its temperature is increased is due to the increase in

A the drift velocity of its electrons
B its amplitude of lattice vibrations
C its mean free time between collisions
D its number of free electrons per unit volume

Answer: B

A higher temperature increases the amplitude of lattice vibrations, so electrons collide more often and the resistivity rises (B).

Explanation: The mean free time decreases; the number of free electrons hardly changes.

6. A wire of radius r which is made of a material of resistivity ρ has a resistance R. If one-tenth of its outer radius is oxidised, the oxidised portion has resistivity five times that of the original material, what is the resistance of the wire?

A 27.53 R B 6.00 R C 1.23 R D 1.18 R

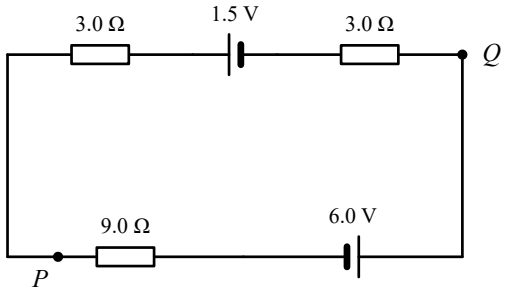
Answer: D

The core (radius $0.9r$, area $0.81A$, resistivity ρ) and the oxidised shell (area $0.19A$, resistivity 5ρ) are in parallel.

$$R_{\text{core}} = \frac{R}{0.81},\ R_{\text{shell}} = \frac{5R}{0.19};\ \frac{1}{R'} = \frac{0.81}{R} + \frac{0.19}{5R} \Longrightarrow R' = 1.18R$$

Explanation: Conductances add in parallel.

7. A circuit consisting of two batteries connected in series with three resistors is shown in the diagram below.



What is the potential difference between point P and point Q?

A 1.8 V B 2.7 V C 3.3 V D 4.5 V

Answer: C

The options assume the two cells oppose each other: net e.m.f. = 6.0 − 1.5 = 4.5 V.

$$I = \frac{4.5}{3.0 + 3.0 + 9.0} = 0.30 \text{ A}$$

$V_{PQ} = 6.0 - (0.30)(9.0) = 3.3$ V (check: $(0.30)(6.0) + 1.5 = 3.3$ V)

Explanation: With the cell symbols exactly as printed the cells would aid and V_{PQ} would be 1.5 V, which is not an option.

8. A voltmeter has a resistance of 1000 Ω and a full scale deflection of 2.50 V. If the voltmeter is to be used to measure a maximum voltage of 10.0 V, a resistor of resistance

A 3000 Ω must be connected in series to the voltmeter
B 3000 Ω must be connected in parallel to the voltmeter
C 4000 Ω must be connected in series to the voltmeter
D 4000 Ω must be connected in parallel to the voltmeter

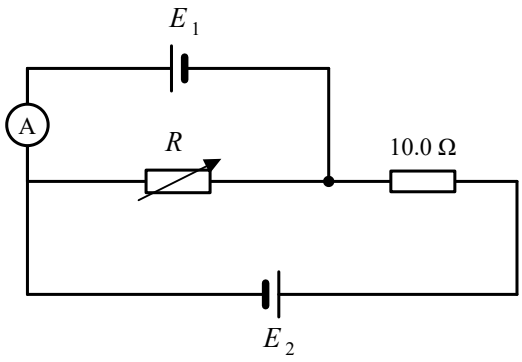
Answer: A

$$\text{Total resistance needed} = \frac{10.0}{2.50 \times 10^{-3}} = 4000\ \Omega \left(\text{full-scale current} = \frac{2.50}{1000} = 2.50 \text{ mA}\right)$$

A 3000 Ω resistor in series (A).

Explanation: A multiplier is always in series with the meter.

9. Two batteries of e.m.f. E_1 and E_2 and a rheostat R are connected in a circuit are shown in the diagram below.



The rheostat is adjusted until the reading of the ammeter becomes zero. If R is 4.0 Ω and E_2 is 2.0 V, what is the value of E_1?

A 0.57 V B 0.80 V C 1.43 V D 2.00 V

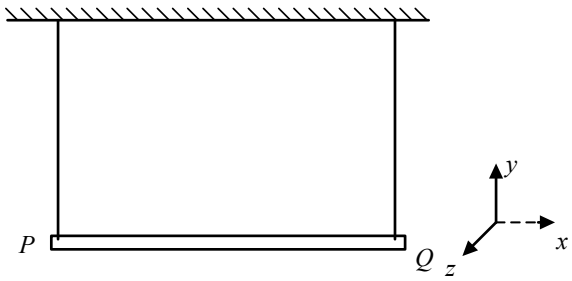
Answer: A

With zero current through E_1, its e.m.f. equals the p.d. across R, which is in series with the 10.0 Ω resistor across E_2.

$$E_1 = 2.0 \times \frac{4.0}{4.0 + 10.0} = 0.57 \text{ V}$$

Explanation: This is the balance condition of a potentiometer.

10. A horizontal rod PQ suspended by wires at both of its ends in a region of magnetic field is shown in the diagram below.



If a current is allowed to flow from Q to P, what is the direction of the magnetic field so that the tension in both wires are reduced?

A Negative-y B Positive-y C Negative-z D Positive-z

Answer: D

The tension is reduced if the magnetic force is upwards ($+y$). With the current along $-x$, $\mathbf{F} = I\mathbf{L} \times \mathbf{B}$ is along $+y$ when $\mathbf{B}$ is along $+z$ (D).

Explanation: $(-x) \times (+z) = +y$.

11. A charged particle travels in a straight line through a region with uniform electric field and magnetic field perpendicular to each other. The electric field strength and the magnetic flux density are 80.0 kV m^{-1} and 0.4 T respectively. What is the speed of the charged particle so that it continues to travel in a straight line?

A 5.0×10^{-6} m s^{-1} B 2.0×10^{2} m s^{-1} C 3.2×10^{4} m s^{-1} D 2.0×10^{5} m s^{-1}

Answer: D

$$qE = qvB \Longrightarrow v = \frac{E}{B} = \frac{80.0 \times 10^{3}}{0.4} = 2.0 \times 10^{5} \text{ m s}^{-1}$$

Explanation: This is independent of the charge and mass.

12. When power is supplied to an electric motor, the motor armature rotates in a magnetic field and an induced e.m.f. is generated. Which statement about the motor is **not** true?

A The induced e.m.f. opposes the e.m.f. supplied.
B The induced e.m.f. is greater than the e.m.f. supplied.
C The induced e.m.f. is maximum when the armature rotates with a maximum speed.
D The different voltage across the armature is smaller than the e.m.f. supplied.

Answer: B

The induced (back) e.m.f. is always less than the supplied e.m.f.; otherwise no current would flow into the motor. Statement B is not true.

Explanation: $V - \varepsilon_b = IR > 0$.

13. A small coil with 100 turns of cross-sectional area 1.26×10^{-3} m^2 is placed coaxially at the centre of a solenoid with 3000 turns and length 2.44 m. What is the mutual inductance between the solenoid and the coil?

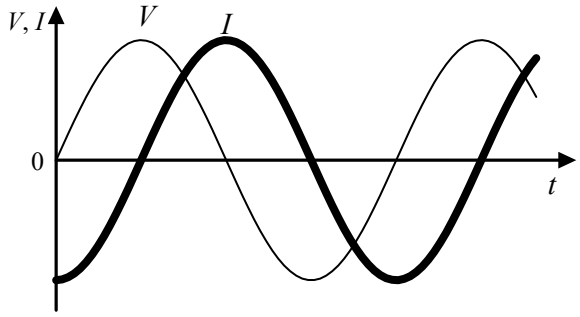
A 6.49×10^{-6} H B 9.73×10^{-5} H C 1.95×10^{-4} H D 5.84×10^{-3} H

Answer: C

$$M = \frac{\mu_0 N_1 N_2 A}{l} = \frac{(4\pi \times 10^{-7})(3000)(100)(1.26 \times 10^{-3})}{2.44} = 1.95 \times 10^{-4} \text{ H}$$

Explanation: l is the length of the solenoid.

14. The variation of voltage V and current I with time t when an alternating current flows through a certain electrical component is shown by the graph below.



Which statement is **not** true of the component?

A The component is an inductor.
B The net power dissipated by the component is zero.
C The reactance of the component is independent of the frequency.
D The voltage across the component always leads the current by $\frac{\pi}{2}$ rad.

Answer: C

The current lags the voltage by 90°, so the component is an inductor (A true) with zero net power (B true) and V leading I by $\pi/2$ (D true).

Its reactance $X_L = 2\pi fL$ depends on frequency, so C is not true.

Explanation: Only a resistor's opposition is independent of frequency.

15. An a.c. supply of frequency f is connected across a capacitor of capacitance C. If the peak voltage of the supply is V_0, what is the maximum current that flows through the capacitor?

A πfCV_0 B $\sqrt{2}\,\pi fCV_0$ C $2\pi fCV_0$ D $2\sqrt{2}\,\pi fCV_0$

Answer: C

$$I_0 = \frac{V_0}{X_C} = V_0(2\pi fC) = 2\pi fCV_0$$

Explanation: Peak voltage over reactance gives peak current.

Section B [15 marks]

16. (a) Define *self-inductance*. [2]

Answer

The self-inductance of a coil is the ratio of the e.m.f. induced in it to the rate of change of current in it — 1

$L = -\frac{\varepsilon}{\mathrm{d}I/\mathrm{d}t}$ (equivalently the flux linkage per unit current, $L = N\Phi/I$) — 1

2

Explanation: Unit: henry (H) = V s A^{-1}.

(b) A constant e.m.f. of 1.04 V is induced by a changing current in a coil of inductance 0.26 H. The initial current is 13.0 A. Calculate, after 2.0 s of supplying the current,

(i) the current in the coil, [3]

Answer

$\frac{dI}{dt} = \frac{\varepsilon}{L} = \frac{1.04}{0.26} = 4.0 \text{ A s}^{-1}$ 1

Change in 2.0 s = (4.0)(2.0) = 8.0 A 1

I = 13.0 − 8.0 = 5.0 A if the current is decreasing (or 21.0 A if increasing) 1

3

Explanation: A constant e.m.f. means the current changes at a constant rate.

(ii) the magnetic flux linkage in the coil. [2]

Answer

$N\Phi = LI = (0.26)(5.0)$ 1

$N\Phi$ = 1.3 Wb (or 5.5 Wb for 21.0 A) 1

2

Total for question **7**

Explanation: Flux linkage is proportional to the current.

17. (a) Derive an expression for the reactance X_L of a pure inductor in terms of its inductance L and the frequency f of a source of current supplied.
[Assume that the expression of the source of current supplied is $I = I_0 \sin 2\pi ft$.] [4]

Answer

$V = L\frac{dI}{dt}$ with $I = I_0 \sin 2\pi ft$ 1

$V = 2\pi fLI_0 \cos 2\pi ft$ 1

Peak voltage $V_0 = 2\pi fLI_0$ 1

$X_L = \frac{V_0}{I_0} = 2\pi fL$ 1

4

Explanation: The voltage leads the current by 90°.

(b) An a.c. circuit contains a pure inductor of inductance 0.470 mH. It is operated by a power supply with peak voltages of 120 V and a frequency of 90.3 MHz. Determine

(i) the peak current of the inductor, [2]

Answer

$X_L = 2\pi(90.3 \times 10^6)(0.470 \times 10^{-3}) = 2.67 \times 10^5\ \Omega$ 1

$I_0 = \dfrac{120}{2.67 \times 10^5} = 4.50 \times 10^{-4}\ \text{A}$ 1

2

Explanation: At this very high frequency the reactance is huge.

(ii) the peak power in the inductor. [2]

Answer

$p = VI = V_0 I_0 \sin \omega t \cos \omega t = \frac{1}{2} V_0 I_0 \sin 2\omega t$ 1

Peak power $= \frac{1}{2}(120)(4.50 \times 10^{-4}) = 0.027$ W 1

2

Total for question **8**

Explanation: The average power in a pure inductor is zero.

Section C [30 marks]

18. (a)

(i) Define the *capacitance* of a capacitor. [1]

Answer

The ratio of the charge on either plate to the p.d. between the plates, $C = \dfrac{Q}{V}$ 1

1

Explanation: Unit: farad.

(ii) A potential difference is applied across an air-filled parallel plate capacitor. State the physical properties of the capacitor that determine the maximum charge stored. If a mica is inserted between the plates, what happens to the capacitance of the capacitor? [3]

Answer

$Q = CV$ with $C = \varepsilon_0 A/d$: the maximum charge depends on the area of overlap of the plates and their separation (for a given p.d.) 1

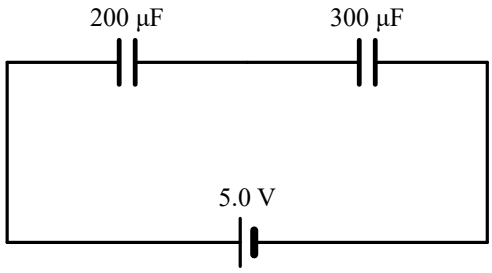
and on the material between the plates (its permittivity) 1

Inserting mica increases the capacitance (by its dielectric constant) 1

3

Explanation: The breakdown field of the dielectric also limits the maximum p.d.

(b) A circuit with two capacitors and a battery is shown in the diagram below.



Determine

(i) the total amount of charge that flows in the circuit, [3]

Answer

$C = \frac{(200)(300)}{200 + 300} = 120\ \mu\text{F}$ 1

$Q = CV = (120 \times 10^{-6})(5.0)$ 1

$Q = 6.0 \times 10^{-4}\ \text{C}$ 1

3

Explanation: In series each capacitor has the same charge.

(ii) the potential difference across the 300 μF capacitor. [2]

Answer

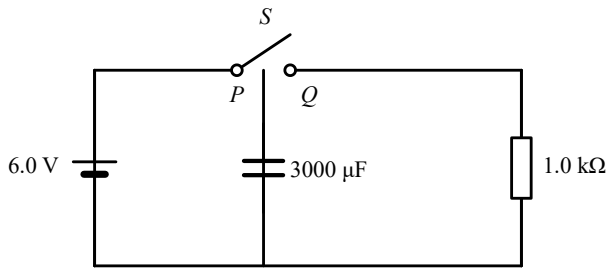
$V = \frac{Q}{C} = \frac{6.0 \times 10^{-4}}{300 \times 10^{-6}}$ 1

$V = 2.0\ \text{V}$ 1

2

Explanation: The 200 μF capacitor has the other 3.0 V.

(c) A circuit consisting of a 6.0 V battery, a 3000 μF capacitor and a 1.0 kΩ resistor is shown in the diagram below.



The switch S is closed at terminal P so that the capacitor is fully charged. Then, it is closed at terminal Q.

(i) Calculate the time constant for the circuit. [2]

Answer

$\tau = RC = (1.0 \times 10^{3})(3000 \times 10^{-6})$ 1

$\tau = 3.0$ s 1

2

Explanation: Ω × F = s.

(ii) Estimate the energy dissipated through the resistor after 2.0 s. [4]

Answer

Initial energy $U_0 = \frac{1}{2}(3000 \times 10^{-6})(6.0)^2 = 5.4 \times 10^{-2}$ J 1

$U \propto Q^2$: $U = U_0 e^{-2t/\tau}$ 1

After 2.0 s: $U = 5.4 \times 10^{-2}\ e^{-4/3} = 1.4 \times 10^{-2}$ J 1

Energy dissipated = $5.4 \times 10^{-2} - 1.4 \times 10^{-2} = 4.0 \times 10^{-2}$ J 1

4

Total for question **15**

Explanation: The energy lost by the capacitor is dissipated as heat in the resistor.

19. (a) An electric iron is labelled 240 V; 1.2 kW. What do you understand by the labelling? [2]

Answer

When connected to a 240 V supply, the iron works normally 1

and converts electrical energy at a rate of 1.2 kW (1200 J every second) 1

2

Explanation: Its operating current is 5.0 A and its resistance 48 Ω.

(b)

(i) Define *drift velocity*. [2]

Answer

The average velocity of the free charge carriers (electrons) along a conductor — 1

when an electric field (p.d.) is applied across it — 1

2

Explanation: It is superimposed on the random motion.

(ii) Copper has a molar mass of 63.5×10^{-3} kg and a density of 8900 kg m^{-3}. Assuming that each copper atom contribute one free electron, calculate the drift velocity of the electrons in a copper wire of radius 1.0 mm carrying a current of 0.15 A. [6]

Answer

Moles per $m^3 = \dfrac{8900}{63.5 \times 10^{-3}} = 1.40 \times 10^5$ mol m^{-3} — 1

$n = (1.40 \times 10^5)(6.02 \times 10^{23}) = 8.44 \times 10^{28}$ m^{-3} — 1

$A = \pi(1.0 \times 10^{-3})^2 = 3.14 \times 10^{-6}$ m^2 — 1

$I = nAev$ — 1

$v = \dfrac{0.15}{(8.44 \times 10^{28})(3.14 \times 10^{-6})(1.60 \times 10^{-19})}$ — 1

$v = 3.54 \times 10^{-6}$ m s^{-1} — 1

6

Explanation: One free electron per copper atom.

(c) A copper conducting rod carries a current of 15.0 A. The cross-sectional area of the copper rod is a square of side 4.0 mm and its length is 60 m. Copper has a resistivity of 1.69×10^{-8} Ω m. Calculate

(i) the density of the current in the copper rod, [3]

Answer

$A = (4.0 \times 10^{-3})^2 = 1.6 \times 10^{-5}$ m^2 — 1

$J = \dfrac{I}{A} = \dfrac{15.0}{1.6 \times 10^{-5}}$ — 1

$J = 9.4 \times 10^5$ A m^{-2} — 1

3

Explanation: Square cross-section: area = side^2.

(ii) the resistance of the copper rod. [2]

Answer

$R = \frac{\rho l}{A} = \frac{(1.69 \times 10^{-8})(60)}{1.6 \times 10^{-5}}$ 1

$R = 6.34 \times 10^{-2}\ \Omega$ 1

2

Total for question **15**

Explanation: A thick copper rod has a very small resistance.

20. (a) Define *magnetic flux density*. [2]

Answer

The force per unit current per unit length on a conductor placed perpendicular to the magnetic field 1

$B = \frac{F}{IL \sin \theta}$; unit tesla (T) 1

2

Explanation: Equivalently $F/(qv \sin \theta)$ for a moving charge.

(b) A straight wire of length 2.75 m carries a current of 5.0 A.

(i) Determine the magnitude of the magnetic flux density at a distance 10.0 cm perpendicular to the wire. [2]

Answer

$B = \frac{\mu_0 I}{2\pi d} = \frac{(4\pi \times 10^{-7})(5.0)}{2\pi(0.100)}$ 1

$B = 1.0 \times 10^{-5}$ T 1

2

Explanation: The wire is treated as long and straight.

(ii) If the wire is looped to form a three turns circular coil, calculate the magnetic flux density at the centre of the coil. [3]

Answer

Circumference of each turn $= \frac{2.75}{3}$ m $\Longrightarrow r = \frac{2.75}{6\pi} = 0.146$ m 1

$B = \frac{N\mu_0 I}{2r} = \frac{(3)(4\pi \times 10^{-7})(5.0)}{2(0.146)}$ 1

$B = 6.5 \times 10^{-5}$ T 1

3

Explanation: The whole length of wire forms the three turns.

(c)

(i) Describe Hall effect. [2]

Answer

When a current-carrying conductor is placed in a magnetic field perpendicular to the current, the charge carriers are pushed to one side — 1

producing a p.d. (Hall voltage) across the conductor, perpendicular to both the current and the field — 1

2

Explanation: The sign of the Hall voltage shows the sign of the carriers.

(ii) Derive an expression for the number of charge carriers per unit volume in a conductor in terms of Hall voltage. [4]

Answer

In equilibrium the electric and magnetic forces balance: $e\frac{V_H}{d} = evB$ — 1

$I = nevA$ with $A = dt$ (d = width, t = thickness) — 1

$v = \frac{V_H}{Bd} \Longrightarrow I = ne\frac{V_H}{Bd}dt$ — 1

$n = \frac{BI}{eV_H t}$ — 1

4

Explanation: t is the thickness along the magnetic field.

(iii) A germanium slab of 4.00 mm thick, 4.00 mm wide and 12.0 mm long is placed with the larger surface perpendicular to a uniform magnetic field of 0.80 T. The Hall voltage measured is 25.0 mV when a current of 50.0 mA flows through the length of the slab. Determine the number of charge carriers per unit volume in the germanium slab. [2]

Answer

$n = \frac{(0.80)(50.0 \times 10^{-3})}{(1.60 \times 10^{-19})(25.0 \times 10^{-3})(4.00 \times 10^{-3})}$ — 1

$n = 2.5 \times 10^{21}\ \text{m}^{-3}$ — 1

2

Total for question **15**

Explanation: Semiconductors have far fewer carriers than metals.